

DRAKE FIELD, AR, USA

WMO: 723445

Lat: 36.010N

Lon: 94.169W

Elev: 1251

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	16.1	1.0	6.1	14.3	6.6	8.1	21.0	24.9	55.4	22.2	52.6	5.4	0	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	95.2	74.4	92.1	74.9	90.1	74.6	78.2	88.2	77.2	87.4	76.2	86.1	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	139.2	83.3	74.2	133.9	82.3	73.1	128.7	81.3	42.7	88.5	41.5	87.3	40.4	86.3	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.9	17.0	3.1	98.1	6.1	4.0	-1.3	100.9	-4.9	103.2	-8.3	105.5	-12.7	108.3		
			2.2	79.9	5.9	1.4	-2.0	80.9	-5.5	81.7	-8.8	82.5	-13.1	83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	36.8	40.8	48.7	57.7	65.9	74.0	78.1	77.4	69.6	58.7	48.1	39.1
	DBStd	16.78	11.12	10.97	10.20	8.43	7.31	5.02	4.67	5.12	6.98	8.53	9.49	9.80
	HDD50	1430	431	294	147	26	2	0	0	0	0	22	147	361
	HDD65	3975	876	679	510	244	80	4	0	0	35	231	512	804
	CDD50	4350	21	36	107	258	495	720	870	850	590	291	89	23
	CDD65	1415	0	0	5	26	107	274	405	386	174	35	3	0
	CDH74	14090	0	7	44	216	881	2608	4270	3991	1724	333	16	0
	CDH80	5748	0	1	4	27	191	936	1979	1909	645	56	0	0
Wind	WSAvg	6.5	7.5	7.5	7.9	8.5	6.8	5.9	5.0	4.6	4.5	5.9	7.1	6.9
Precipitation	PrecAvg	43.2	2.2	2.6	3.9	4.3	5.1	4.8	2.9	3.8	4.5	3.6	3.5	2.8
	PrecMax	69.0	5.8	6.7	8.7	8.0	10.5	8.3	7.1	9.1	10.4	9.6	9.6	6.9
	PrecMin	21.0	0.5	1.2	1.0	1.3	1.3	1.7	0.3	0.5	0.7	0.0	0.9	0.7
	PrecStd	10.1	1.5	1.4	2.1	1.9	2.4	1.9	2.2	2.3	2.7	3.0	2.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.8	72.7	78.8	83.0	88.0	93.4	98.9	99.8	94.6	84.5	76.4	69.1
		MCWB	57.6	58.3	61.9	65.7	71.6	74.3	73.9	74.0	72.6	69.8	63.6	60.1
	2%	DB	64.2	67.2	73.9	79.4	84.3	90.3	94.6	96.3	90.0	81.4	71.7	64.4
		MCWB	56.3	56.5	60.1	63.7	70.4	75.3	75.0	74.2	72.2	67.6	60.4	58.1
	5%	DB	60.5	63.6	70.3	76.0	81.9	87.9	91.5	92.7	87.2	77.9	68.1	61.0
		MCWB	53.0	54.8	58.0	62.3	69.8	74.4	75.8	74.3	71.9	65.2	59.3	53.8
	10%	DB	55.3	60.2	66.1	73.0	79.3	85.5	89.9	90.0	83.6	74.1	64.3	55.9
		MCWB	48.2	52.4	56.0	61.1	68.4	73.4	75.7	74.2	70.5	64.1	56.9	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	63.1	65.2	69.3	75.3	78.6	79.9	79.5	76.7	73.0	67.2	63.1
		MCDB	66.0	68.0	74.1	78.2	82.5	87.8	89.2	88.9	87.5	81.4	73.8	66.5
	2%	WB	58.9	60.1	62.8	67.3	73.3	77.1	78.5	77.8	75.1	70.9	63.8	59.3
		MCDB	63.2	65.1	70.9	75.1	81.1	86.8	88.4	88.3	85.6	78.0	68.1	64.2
	5%	WB	54.3	56.9	60.3	65.7	71.5	75.7	77.6	76.6	73.7	68.5	61.1	55.5
		MCDB	58.1	62.3	67.8	73.1	79.1	84.6	87.7	87.4	83.0	74.3	66.4	59.7
	10%	WB	49.4	52.8	57.9	63.8	69.9	74.6	76.6	75.3	72.2	66.0	58.3	51.3
		MCDB	55.4	58.4	64.9	70.7	77.5	83.1	86.3	86.1	80.4	72.1	63.4	54.8
Mean Daily Temperature Range	5% DB	MDBR	20.7	21.3	22.5	23.0	20.6	19.8	20.5	21.9	23.0	23.5	22.0	20.0
		MCDBR	26.0	26.6	26.8	26.1	22.2	22.2	23.1	25.8	25.8	26.5	25.5	25.5
		MCWBR	17.8	17.9	15.7	14.1	11.1	9.6	8.2	8.3	10.5	13.9	16.0	18.2
	5% WB	MCDBR	21.0	22.3	22.0	20.1	18.8	19.6	20.1	20.8	21.4	20.2	20.6	21.1
		MCWBR	17.4	17.5	15.1	13.1	10.6	9.6	8.9	8.4	9.8	12.1	15.8	17.3
Clear-Sky Solar Irradiance	taub	0.304	0.307	0.341	0.378	0.417	0.425	0.447	0.446	0.388	0.345	0.315	0.293	
	taud	2.493	2.492	2.427	2.330	2.272	2.273	2.265	2.242	2.377	2.484	2.521	2.541	
	Ebn at Noon	285	298	294	287	276	272	266	264	275	279	279	282	
	Edh at Noon	26	29	34	40	43	43	43	43	36	29	25	23	
All-Sky Solar Radiation	RadAvg	784	947	1282	1612	1781	2030	2014	1835	1566	1171	843	671	
	RadStd	71	129	77	93	155	189	123	124	118	152	95	85	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page